Q1

1. A mountaineer arrives at a location X, which is 3000 m above sea level. At X, the melting point of ice is 0 °C and the boiling point of water is 90 °C. The mountaineer needs to heat 0.50 kg of melting ice to its boiling point using a heater.

(a) Estimate the energy required. (2 marks)

Given: specific latent heat of fusion of ice = 3.34×10^5 J kg^{-1}
specific heat capacity of water = 4200 J kg^{-1} °C^{-1}

Q2

(b) The power output of the heater is 2 kW. Estimate the time required. (2 marks)

(c) Explain why the temperature of water remains at the boiling point after the water starts to boil, even when the heater is still on. (1 mark)

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*2. A sealed syringe contains neon gas of volume 5.0 cm^3 and pressure 1.0×10^5 Pa. The gas is then compressed slowly to 2.0 cm^3 while the temperature remains at 27 °C. Take the neon gas as an ideal gas.

(a) Find the pressure of the gas after being compressed. (2 marks)

(b) (i) Find the average kinetic energy of the gas molecules. (2 marks)

(ii) The mass of one mole of neon gas is 0.020 kg. Find the root-mean-square speed of the gas molecules. (2 marks)

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Q3

3. A uniform metal rod PQ of weight 12 N and length 0.80 m is hinged smoothly to the edge of a table at P as shown in Figure 3.1. One end of a light inextensible string is attached to Q and the other end is fixed to a heavy box placed on the table such that the rod is kept horizontal. The string makes 36° with the rod.

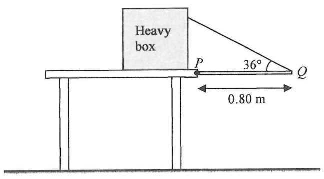

Figure 3.1

(a) Find the tension of the string (2 marks)

The string is cut. The rod falls, strikes a small ball resting on the floor and exerts a horizontal force on it towards the left, as shown in Figure 3.2. After the collision, the ball moves to the left with speed 1.6 m s^{-1}.

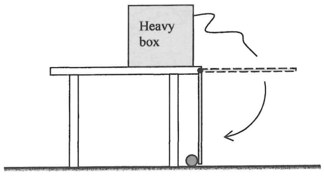

Figure 3.2

(b) (i) If the mass of the ball is 0.05 kg and the collision time is 0.008 s, find the magnitude of the average force acting on the ball by the rod. (2 marks)

(ii) Explain why the momentum of the ball is not conserved before and after the collision. (1 mark)

Q4

4. Figure 4.1(a) shows a smooth track *ABCD* consisting of a sloping part *AB* and a horizontal semicircular part *BCD* fixed on a table. Point *A* is at 0.80 m above the table. The top view of the track is shown in Figure 4.1(b).

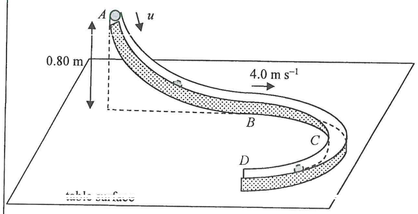

Figure 4.1(a)

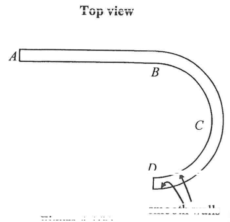

Figure 4.1(b)

A small sphere is projected along the track with speed u at A. It moves along the track and passes B at 4.0 m s^{-1}. Neglect air resistance. ($g = 9.81 \text{ m s}^{-2}$)

(a) By using the law of conservation of energy, find u. (2 marks)

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*(b) The sphere then moves along *BCD* with constant speed 4.0 m s^{-1}. The radius of *BCD* is 0.20 m.

(i) Explain why, even though the speed of the sphere remains constant, its acceleration is not zero during the motion described above. (1 mark)

(ii) Estimate the magnitude of the acceleration of the sphere. (1 mark)

(iii) What force provides the centripetal force for the sphere ? (1 mark)

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Q5

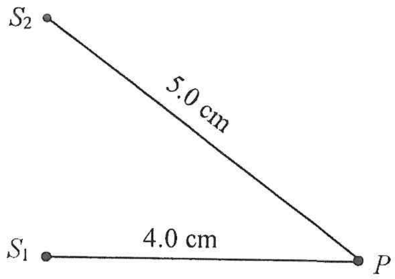
5. In a ripple tank, circular water waves are produced by two dippers S_1 and S_2 connected to a vibrator. The vibrator works in a way that S_1 and S_2 always vibrate in opposite directions as shown in Figure 5.1. The frequency of the vibration is 10 Hz and the wavelength λ of the waves is 2.0 cm.

Figure 5.1

(a) Find the wave speed of the waves. (1 mark)

(b) (i)

Figure 5.2



P is a point at the water surface as shown in Figure 5.2. Express the path difference at P from S_1 and S_2 in terms of λ. (2 marks)

(ii) Explain briefly what type of interference occurs at P. (2 marks)

(c) If the depth of the water in the ripple tank is increased while other conditions remain unchanged, deduce how the wavelength of the waves will be affected. (2 marks)

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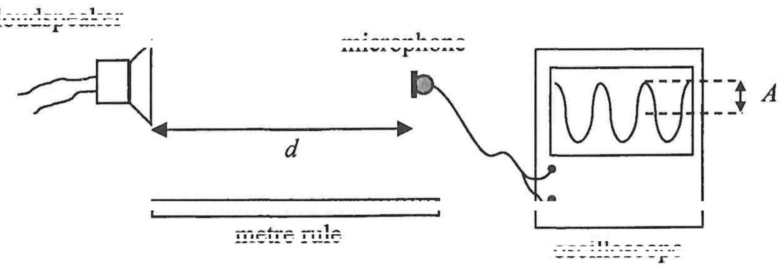
Q6

6. A student uses the set-up in the figure below to investigate the relation between the loudness of sound and the distance from the source of sound. The distance between the loudspeaker and the microphone is d. The loudness of the sound relates to the amplitude A of the oscilloscope trace.

The student hypothesises that A is inversely proportional to d^2 ($A \propto \frac{1}{d^2}$).



Describe

~~(I) the procedure of the experiment;~~

(II) how the data should be analysed using a graphical method to verify whether the hypothesis ($A \propto \frac{1}{d^2}$) is valid.

(5 marks)

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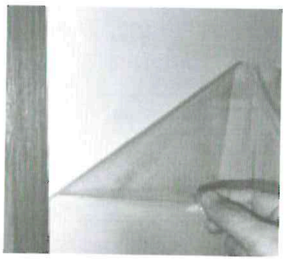
Q7

7. Read the following passage about **electrostatic adhesive wallpaper** and answer the questions that follow.

Electrostatic adhesive wallpaper is a wall covering that uses **static electricity** for adhesion, without the need for traditional glues or pastes. The wallpaper is typically made from a material that can retain electric charges on its surface and is charged through friction during production, allowing it to adhere directly to smooth, clean surfaces like painted walls or glass.



The wallpaper can be easily applied or removed. After removal, the wallpaper does not leave residue or damage the underlying surface. Therefore, the wallpaper is especially suitable for quick updates and temporary setups. Another advantage of the wallpaper is its reusability. It preserves its adhesive properties after being removed and can be reapplied.

(a) Describe how the electrostatic adhesive wallpaper can adhere to a painted wall that is initially neutral. (2 marks)

(b) Explain briefly why the wallpaper preserves its adhesive properties after being removed. (2 marks)

(c) The wallpaper **does not** adhere well to the wall in a humid environment. Explain briefly. (1 mark)

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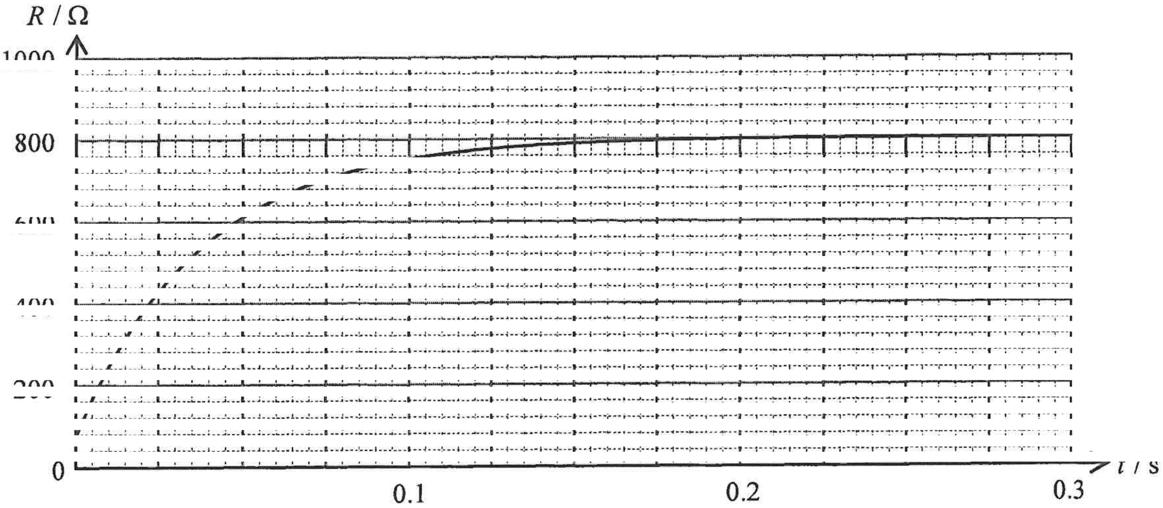
Q8

8. A tungsten filament light bulb is connected to a 220-V supply. The following graph shows the variation of the resistance R of the bulb with time t after it is switched on.



(a) The bulb works at its rated value after R becomes stable. Estimate the rated power of the bulb. (2 marks)

(b) Explain why R initially increases and becomes stable after a period of time. (3 marks)

(c) The filament of the bulb has a length of 0.56 m and a uniform cross-section of diameter 0.022 mm. Find the resistivity of the filament at $t = 0$ s. (2 marks)

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Q9

*9. A long, straight wire carries a sinusoidal a.c. of r.m.s. current I_{rms} of 5 A. A search coil with 500 turns, each having an area of 4.0×10^{-4} m², is placed 0.20 m away from the wire, as shown below. The wire and the search coil are in the same plane.

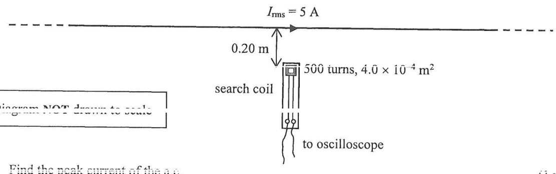

Diagram NOT drawn to scale

(a) Find the peak current of the a.c. (1 mark)

(b) Estimate the maximum magnetic flux linkage produced by the wire through the search coil. (3 marks)

(c) A student claims that the search coil can be used to measure the resultant magnetic field produced by the Earth and the wire. Explain briefly whether this claim is correct. (2 marks)

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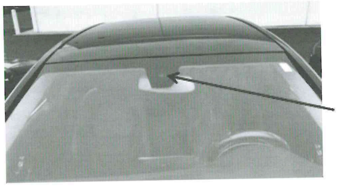
It senses rainwater on the glass and

Q10

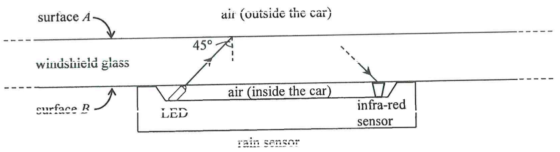
10. Figure 10.1 shows a rain sensor mounted on the windshield glass of a car. It senses rainwater on the glass and automatically turns on the windshield wipers.

Figure 10.1



As shown in Figure 10.2, the rain sensor is composed of a light emitting diode (LED) and an infra-red sensor. The LED emits an infra-red signal and the infra-red sensor measures the intensity of the signal. When the intensity changes, the wipers are activated. An infra-red ray from the LED incident at surface A at 45°, and the infra-red ray finally received by the infra-red sensor are drawn in Figure 10.2. Assume surface A is parallel to surface B. It is given that the critical angle of the glass for infra-red is 44.0°.

Figure 10.2



(a) Explain how the infra-red ray in Figure 10.2 travels from the LED to the infra-red sensor. (4 marks)

(b) When it rains, some raindrops settle on the glass surface, as illustrated in Figure 10.3(a). Figure 10.3(b) shows an enlarged view of the region where the infra-red ray is incident on the glass-water interface.

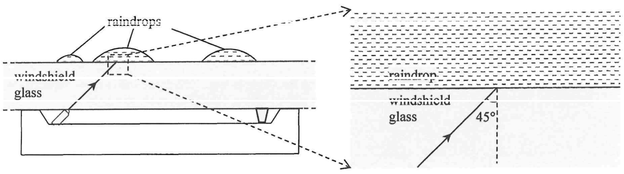


Figure 10.3(a) Figure 10.3(b)

(i) Given that the refractive index of water for the infra-red is 1.26, find the critical angle of the glass-water interface for the infra-red ray. (3 marks)

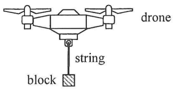
(ii) Sketch in Figure 10.3(b) how the infra-red ray propagates after being incident on the glass-water interface. (2 marks)

(iii) The wipers are activated when the intensity of the signal received by the infra-red sensor changes. Explain briefly how the intensity changes. (2 marks)

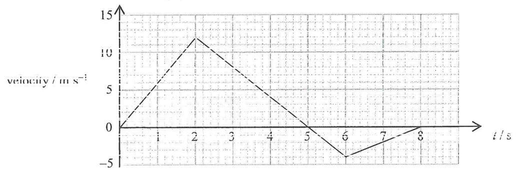
Q11

11. A drone carries a 0.50-kg block by connecting it with a light and inextensible string as shown in Figure 11.1. The size of the block is negligible. ($g = 9.81$ m s^{-2})

Figure 11.1

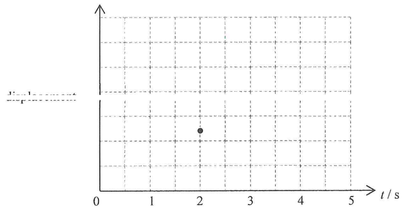


(a) At time $t = 0$ s, the block is at rest on the ground and the string is taut. The drone then moves vertically with the block. The velocity-time graph of the block is shown below, taking upwards as positive.



(i) Find the maximum height that the block can reach from $t = 0$ to 8 s. (2 marks)

(ii) Sketch the displacement-time graph of the block from $t = 0$ to 5 s. The displacement at $t = 2$ s is marked. Labelling of values is not required and the displacement at $t = 0$ s is taken to be zero. (2 marks)



(iii) Find the tension of the string at $t = 5$ s. (3 marks)

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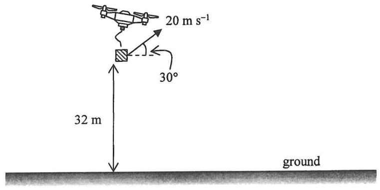
*(b) The drone now flies upward and forward. When the block is 32 m above the ground, the string breaks. Figure 11.2 shows the moment immediately after the break, where the block has a velocity of 20 m s^{-1} at an angle of 30° above the horizontal. Neglect air resistance.

Figure 11.2



(i) Draw an arrow in the figure below to represent the direction of the acceleration of the block at that moment. (1 mark)

(ii) Find the horizontal displacement of the block from that moment until just before it reaches the ground. (3 marks)

(iii) Describe the energy conversion of the block from that moment until just before it reaches the ground. (2 marks)

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Q12

12. (a) Bismuth-213 (Bi-213) is a radionuclide currently undergoing clinical trials for cancer treatment. It is prepared in liquid and injected into the human body, where it targets and destroys cancer cells. Figure 12.1 shows a decay series of Bi-213.

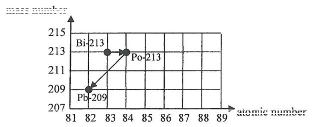

Figure 12.1

(i) Write the nuclear equations corresponding to each of these two decay processes. (2 marks)

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*(ii) The half-life of Bi-213 is 46 minutes. At time $t = 0$ s, its number of undecayed nuclei is 8.0×10^{11}. Find its activity (in Bq) at $t = 0$ s. (2 marks)

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*(iii) Find the percentage decrease in the activity after 200 minutes. (2 marks)

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(b) In hospital, technicians wear protective gloves to handle liquid radioactive materials, such as those containing Bi-213. To monitor radiation exposure, they wear a specialised ring on their finger, as shown in Figure 12.2(a). Each ring contains a small chip in a plastic enclosure (Figure 12.2(b)), which detects and records radiation doses. The rings are collected every three months to monitor the radiation exposure of the technicians.

Figure 12.2(a)

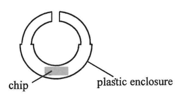

Figure 12.2(b)

(i) Compared to placing it in a badge worn on the chest, what is the advantage of placing the chip in the ring worn on the hand ? (1 mark)

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(ii) The ring should be worn under the gloves. Explain briefly. (1 mark)

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(iii) Which type(s) of radiation (α, β or γ) can reach the chip of the ring ? (1 mark)

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END OF PAPER

Sources of materials used in this paper will be acknowledged in the *HKDSE Question Papers* booklet published by the Hong Kong Examinations and Assessment Authority at a later stage.

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